Speed

What a speed measures, how it is written, and how fast things actually go.

LESSON 117–119 | 3 × 40 MIN | MATEMATIKA 6, §24 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

What a speed is	25 min
Finding a speed	30 min
Comparing speeds	35 min
Is it reasonable?	25 min
Homework	5 min

LEARNING OBJECTIVES

- Define speed as distance per unit of time.
- Compare speeds given in different units.
- Read a speed from a table or a description.
- Judge whether a speed is reasonable.

TEXTBOOK

National	pp. 339–348
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.5

01 Explanation

What a speed is

$$v = \frac{s}{t}$$

A speed of *60* km/h means *60* km in every hour — and therefore *30* km in half an hour, *1* km per minute, and *16.7* m each second.

SPEED	IN ONE HOUR	IN ONE MINUTE
60 km/h	*60 km*	*1 km*
90 km/h	*90 km*	*1.5 km*
5 km/h	*5 km*	*83 m*

A SPEED IS A RATE: SOMETHING PER SOMETHING

Price per kilogram, pupils per class, kilometres per hour — all are rates, and all are found by dividing one quantity by another.

Finding a speed

JOURNEY	WORKING	SPEED
180 km in *3* h	$180 \div 3$	*60 km/h*
240 km in *4* h	$240 \div 4$	*60 km/h*
100 m in *12.5* s	$100 \div 12.5$	*8 m/s*
9 km in *40* min	$9 \div \frac{2}{3}$	*13.5 km/h*

DIVIDE THE DISTANCE BY THE TIME, NEVER THE OTHER WAY ROUND

$3 \div 180$ gives *0.017*, which is hours per kilometre — a real quantity, but not the one asked for. The units tell you which division you have done.

Comparing speeds

Two speeds can only be compared in the same units.

A	B	IN ONE UNIT	FASTER
72 km/h	*18 m/s*	*20* and *18* m/s	A
54 km/h	*16 m/s*	*15* and *16* m/s	B
4 km/h	*70 m/min*	*66.7* and *70* m/min	B

CONVERT FIRST, COMPARE SECOND

The larger number is not the faster speed unless the units match. This is the same rule as comparing *50* cm with *2* m.

Is it reasonable?

MOVER	TYPICAL SPEED
a walker	*5 km/h*
a cyclist	*15–20 km/h*
a car in a town	*40–60 km/h*
a train	*100–160 km/h*
an aeroplane	*800–900 km/h*
sound in air	*340 m/s*

AN ANSWER OF *600* KM/H FOR A CYCLIST IS A SIGNAL, NOT A RESULT

Comparing an answer with the table catches unit errors and misplaced decimal points before the working is even reread.

02 Worked examples

EXAMPLE 1 **A runner covers *100* m in *12.5* seconds. Find the speed.**

1. $v = \frac{s}{t} = \frac{100}{12.5}$
2. $= 8$ m/s.
3. In km/h that is *28.8* — reasonable for a sprinter.

ANSWER

8 m/s

EXAMPLE 2 **Which is faster, 72 km/h or 18 m/s?**

1. $72 \div 3.6 = 20$ m/s.
2. $20 > 18$.
3. The first is faster.

ANSWER

72 km/h

EXAMPLE 3 **A cyclist covers 9 km in 40 minutes. Find the speed in km/h.**

1. 40 min $= \dfrac{2}{3}$ h.
2. $9 \div \dfrac{2}{3} = 9 \cdot \dfrac{3}{2}$
3. $= 13.5$ km/h.

   Reasonable for a cyclist ✓

ANSWER

13.5 km/h

03 Interactive model

Have the class estimate the speed of a walk down the corridor, then measure it; being within 20% is a good result and makes the units real.

Speeds and their units

180 km in *3* h:

- *60 km/h*
- *540 km/h*
- *0.017*
- *183*

100 m in *12.5* s:

- *8 m/s*
- *12.5 m/s*
- *0.125 m/s*
- *1250 m/s*

9 km in *40* min:

- *13.5 km/h*
- *0.225 km/h*
- *360 km/h*
- *22.5 km/h*

Which is faster, 72 km/h or 18 m/s?

- 72 km/h
- 18 m/s
- equal
- cannot say

Which is faster, 54 km/h or 16 m/s?

- 54 km/h
- 16 m/s
- equal
- cannot say

A walker's speed is about:

- 1 km/h
- 5 km/h
- 20 km/h
- 60 km/h

Sound travels at about:

- 34 m/s
- 340 m/s
- 3400 m/s
- 34 km/h

600 km/h for a cyclist means:

- a fast cyclist
- an error somewhere
- a downhill
- a record

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Speed	Tezlik	Скорость
Per hour	Bir soatda	В час
Per second	Bir sekundda	В секунду
Uniform speed	Bir tekis tezlik	Равномерная скорость
To compare	Taqqoslash	Сравнить
Reasonable	Maqbul	Разумный
Rate	Me’yor	Норма
Unit of speed	Tezlik birligi	Единица скорости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Speed is:

- distance times time
- distance over time
- time over distance
- distance plus time

240 km in 4 h is:

- 60 km/h
- 960 km/h
- 0.017 km/h
- 244 km/h

To compare km/h with m/s you:

- add them
- convert one
- take the larger number
- divide them

4 km/h in metres per minute is about:

- 40
- 67
- 240
- 4

A train's speed is about:

- 15 km/h
- 60 km/h
- 120 km/h
- 800 km/h

Hours per kilometre is:

- a speed
- the reciprocal of a speed
- nonsense
- the same thing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 180 km in 3 h

   ANSWER 60 *km/h*

2. 240 km in 4 h

   ANSWER 60 *km/h*

3. 100 m in 12.5 s

   ANSWER 8 *m/s*

4. 60 km in 1 h

   ANSWER 60 *km/h*

5. 20 m in 4 s

   ANSWER 5 *m/s*

6. A walker’s speed

   ANSWER about 5 *km/h*

7. Sound in air

   ANSWER 340 *m/s*

Medium · the standard question

7 PROBLEMS

1. 9 km in 40 min

ANSWER 13.5 km/h

2. 3 km in 45 min

ANSWER 4 km/h

3. Faster: 72 km/h or 18 m/s?

ANSWER 72 km/h

4. Faster: 54 km/h or 16 m/s?

ANSWER 16 m/s

5. 4 km/h in m/min

ANSWER 66.7

6. 90 km/h in m/s

ANSWER 25

7. A cyclist at 600 km/h means

ANSWER an error somewhere

Hard · stretch and reason

7 PROBLEMS

1. A train covers 420 km in $3\ h\ 30\ min$

   ANSWER 120 km/h

2. An aeroplane covers $2\ 700$ km in 3 h

   ANSWER 900 km/h

3. Light takes 8 minutes from the Sun, 150 million km away: the speed

   ANSWER About $312\ 500$ km/s

4. A 400 m lap in 50 s

   ANSWER 8 m/s, or 28.8 km/h

5. Which is faster: 1 km per minute or 60 km/h?

   ANSWER Equal

6. A snail at 1 cm/s in km/h

   ANSWER 0.036

7. Why compare an answer with a table of typical speeds?

   ANSWER It catches unit errors at once

07 Homework

Homework — 5 tasks

Check every answer against a typical speed for that kind of mover.

1. A car covers 330 km in 5 hours. Find its speed.
2. A sprinter runs 200 m in 25 seconds. Find the speed in m/s and in km/h.
3. A walker covers 6 km in $1\ h\ 20\ min$. Find the speed.
4. Which is faster: 90 km/h or 26 m/s?
5. Write down a typical speed for a walker, a car and an aeroplane.